

CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

# Chapter 8: Introduction to Trigonometry

Topic-wise short notes with solved examples

Trigonometry relates the sides of a right-angled triangle to its acute angles. This chapter covers the six ratios, their values for standard angles and the three basic identities.

## 1 Trigonometric ratios

- In a right-angled triangle  $ABC$ , right-angled at  $B$ , take the acute angle  $A$ . Then  $BC$  is the side **opposite** to  $A$ ,  $AB$  is the side **adjacent** to  $A$  and  $AC$  is the **hypotenuse**.

| Ratio    | Definition                                  | Ratio                    | Definition                                  |
|----------|---------------------------------------------|--------------------------|---------------------------------------------|
| $\sin A$ | $\frac{\text{opposite}}{\text{hypotenuse}}$ | $\operatorname{cosec} A$ | $\frac{\text{hypotenuse}}{\text{opposite}}$ |
| $\cos A$ | $\frac{\text{adjacent}}{\text{hypotenuse}}$ | $\sec A$                 | $\frac{\text{hypotenuse}}{\text{adjacent}}$ |
| $\tan A$ | $\frac{\text{opposite}}{\text{adjacent}}$   | $\cot A$                 | $\frac{\text{adjacent}}{\text{opposite}}$   |

- Reciprocal relations:**  $\operatorname{cosec} A = \frac{1}{\sin A}$ ,  $\sec A = \frac{1}{\cos A}$ ,  $\cot A = \frac{1}{\tan A}$ .
- Quotient relations:**  $\tan A = \frac{\sin A}{\cos A}$  and  $\cot A = \frac{\cos A}{\sin A}$ .
- The value of a ratio depends only on the angle, not on the size of the triangle.
- $\sin A$  is one symbol. It does not mean  $\sin \times A$ .

### REMEMBER

The hypotenuse is the longest side, so  $\sin A$  and  $\cos A$  are never greater than 1, and  $\sec A$  and  $\operatorname{cosec} A$  are never less than 1.

## 2 Finding all ratios from one

- Step 1: draw a right-angled triangle and mark two sides using the given ratio, as  $k$  times the numbers.
- Step 2: find the third side with the Pythagoras theorem.
- Step 3: write the required ratios.

**Example 2.1** If  $\sin A = \frac{5}{13}$ , find  $\cos A$  and  $\tan A$ .

SOLUTION

Take opposite side =  $5k$  and hypotenuse =  $13k$ .

Adjacent side =  $\sqrt{(13k)^2 - (5k)^2} = \sqrt{144k^2} = 12k$

$$\cos A = \frac{12}{13} \text{ and } \tan A = \frac{5}{12}$$

**Example 2.2** If  $\tan \theta = \frac{3}{4}$ , evaluate  $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$  and show that it equals  $\cos^2 \theta - \sin^2 \theta$ .

SOLUTION

$$\frac{1 - \frac{9}{16}}{1 + \frac{9}{16}} = \frac{7/16}{25/16} = \frac{7}{25}$$

Opposite =  $3k$ , adjacent =  $4k$ , hypotenuse =  $5k$ , so  $\sin \theta = \frac{3}{5}$  and  $\cos \theta = \frac{4}{5}$

$$\cos^2 \theta - \sin^2 \theta = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$$

Both sides equal  $\frac{7}{25}$ .

### 3 Ratios of standard angles

| Angle | $0^\circ$   | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$  |
|-------|-------------|----------------------|----------------------|----------------------|-------------|
| sin   | 0           | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1           |
| cos   | 1           | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0           |
| tan   | 0           | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | Not defined |
| cosec | Not defined | 2                    | $\sqrt{2}$           | $\frac{2}{\sqrt{3}}$ | 1           |
| sec   | 1           | $\frac{2}{\sqrt{3}}$ | $\sqrt{2}$           | 2                    | Not defined |
| cot   | Not defined | $\sqrt{3}$           | 1                    | $\frac{1}{\sqrt{3}}$ | 0           |

- To remember the sin row:  $\sqrt{\frac{0}{4}}, \sqrt{\frac{1}{4}}, \sqrt{\frac{2}{4}}, \sqrt{\frac{3}{4}}, \sqrt{\frac{4}{4}}$ . The cos row is the sin row in reverse.
- As the angle increases from  $0^\circ$  to  $90^\circ$ , sin increases from 0 to 1 and cos decreases from 1 to 0.

**Example 3.1** Evaluate  $4 \sin^2 60^\circ + 3 \tan^2 30^\circ - 8 \sin 45^\circ \cos 45^\circ$ .

SOLUTION

$$= 4 \times \frac{3}{4} + 3 \times \frac{1}{3} - 8 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$$

$$= 3 + 1 - 4$$

$$0$$

**Example 3.2** If  $\sin(A + B) = 1$  and  $\cos(A - B) = \frac{\sqrt{3}}{2}$ , where  $0^\circ < A + B \leq 90^\circ$  and  $A > B$ , find  $A$  and  $B$ .

SOLUTION

$$\sin(A + B) = 1 = \sin 90^\circ, \text{ so } A + B = 90^\circ$$

$$\cos(A - B) = \frac{\sqrt{3}}{2} = \cos 30^\circ, \text{ so } A - B = 30^\circ$$

$$\text{Adding: } 2A = 120^\circ$$

$$A = 60^\circ \text{ and } B = 30^\circ$$

**Example 3.3** In  $\triangle ABC$ , right-angled at  $B$ ,  $AC = 12$  cm and  $\angle A = 60^\circ$ . Find  $AB$  and  $BC$ .

SOLUTION

$$\cos 60^\circ = \frac{AB}{AC}, \text{ so } AB = 12 \times \frac{1}{2} = 6 \text{ cm}$$

$$\sin 60^\circ = \frac{BC}{AC}, \text{ so } BC = 12 \times \frac{\sqrt{3}}{2} = 6\sqrt{3} \text{ cm}$$

$$AB = 6 \text{ cm and } BC = 6\sqrt{3} \text{ cm}$$

#### REMEMBER

$\sin(A + B)$  is not  $\sin A + \sin B$ . Check with  $A = B = 30^\circ$ :  $\sin 60^\circ = \frac{\sqrt{3}}{2}$ , but  $\sin 30^\circ + \sin 30^\circ = 1$ .

## 4 Trigonometric identities

- An identity is true for every value of the angle for which both sides are defined.

$$\sin^2 A + \cos^2 A = 1 \quad 1 + \tan^2 A = \sec^2 A \quad 1 + \cot^2 A = \operatorname{cosec}^2 A$$

- Each identity can be rearranged:  $\sin^2 A = 1 - \cos^2 A$ ,  $\sec^2 A - \tan^2 A = 1$ ,  $\operatorname{cosec}^2 A - \cot^2 A = 1$ .
- $\sin^2 A$  means  $(\sin A)^2$ .

**Example 4.1** Express  $\cos A$  and  $\tan A$  in terms of  $\sin A$ .

SOLUTION

From  $\sin^2 A + \cos^2 A = 1$ :  $\cos A = \sqrt{1 - \sin^2 A}$

$$\tan A = \frac{\sin A}{\cos A} = \frac{\sin A}{\sqrt{1 - \sin^2 A}}$$

$$\cos A = \sqrt{1 - \sin^2 A} \text{ and } \tan A = \frac{\sin A}{\sqrt{1 - \sin^2 A}}$$

**Example 4.2** Prove that  $(1 + \tan^2 A)(1 - \sin A)(1 + \sin A) = 1$ .

SOLUTION

$$\begin{aligned} \text{LHS} &= \sec^2 A (1 - \sin^2 A) \\ &= \sec^2 A \times \cos^2 A = \frac{1}{\cos^2 A} \times \cos^2 A \\ &= 1 = \text{RHS} \end{aligned}$$

**Example 4.3** Prove that  $\tan A + \cot A = \sec A \operatorname{cosec} A$ .

SOLUTION

$$\begin{aligned} \text{LHS} &= \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} = \frac{\sin^2 A + \cos^2 A}{\sin A \cos A} \\ &= \frac{1}{\sin A \cos A} = \frac{1}{\cos A} \times \frac{1}{\sin A} \\ &= \sec A \operatorname{cosec} A = \text{RHS} \end{aligned}$$

**Example 4.4** Prove that  $\sqrt{\frac{1 - \cos A}{1 + \cos A}} = \operatorname{cosec} A - \cot A$ .

SOLUTION

Multiply the numerator and denominator inside the root by  $(1 - \cos A)$ :

$$\begin{aligned} \text{LHS} &= \sqrt{\frac{(1 - \cos A)^2}{1 - \cos^2 A}} = \sqrt{\frac{(1 - \cos A)^2}{\sin^2 A}} = \frac{1 - \cos A}{\sin A} \\ &= \frac{1}{\sin A} - \frac{\cos A}{\sin A} \\ &= \operatorname{cosec} A - \cot A = \text{RHS} \end{aligned}$$

## 5 How to prove an identity

- Start from the more complicated side and work towards the simpler one.
- If nothing else suggests itself, change every ratio to  $\sin$  and  $\cos$ .
- For a sum of fractions, take the LCM. For  $1 \pm \sin A$  or  $1 \pm \cos A$  in a denominator, multiply by the conjugate.
- Look for  $1 - \sin^2 A$ ,  $1 - \cos^2 A$ ,  $\sec^2 A - 1$  and  $\operatorname{cosec}^2 A - 1$ : each one is a single square.
- Never move terms from one side to the other in a proof. Work on one side only, or simplify both sides separately.

## 6 Common mistakes

- Writing  $\sin A = \frac{\text{opposite}}{\text{adjacent}}$ . That is  $\tan A$ .
- Treating  $\sin^2 A$  as  $\sin A^2$  or  $\sin 2A$ .
- Using  $\tan 90^\circ$  or  $\sec 90^\circ$  as a number. They are not defined.
- Cancelling  $\sin$  from  $\frac{\sin A}{\sin B}$  to get  $\frac{A}{B}$ .